Iteration – Self Learning in Python

1. Consecutive Digit Number

Problem Statement:
Accept a number and check whether every digit is exactly 1 greater than its previous digit.

Example:

Input: 12345
Output: Consecutive Number

Input: 1357
Output: Not a Consecutive Number

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2. Mountain Number

Problem Statement:
A Mountain Number is a number whose digits first increase and then decrease.

Example:

Input: 12321
Output: Mountain Number

Input: 12345
Output: Not a Mountain Number

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3. Longest Increasing Sequence

Problem Statement:
Accept N numbers one by one and find the length of the longest continuous increasing sequence.

Example:

Input:
5 8 10 12 3 4 5 6 1

Output:
Longest Sequence Length = 4

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4. Electricity Bill Simulator

Problem Statement:
Calculate electricity bill using slabs:

0-100 units -> ₹5/unit
101-200 units -> ₹7/unit
Above 200 units -> ₹10/unit

Additionally:

- Add 10% surcharge if bill exceeds ₹5000.

Display final payable amount.

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5. Secret Code Validator

Problem Statement:
A secret code is valid if:

- It contains exactly 6 digits.
- Sum of first 3 digits equals sum of last 3 digits.

Example:

Input: 123321
Output: Valid Code

Input: 123456
Output: Invalid Code

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6. Race Leader Tracker

Problem Statement:
Input lap times of N racers.

Display:

- Fastest racer position
- Slowest racer position
- Difference between fastest and slowest lap time

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7. Number Mirror Check

Problem Statement:
Check whether the left half of a number is identical to the right half.

Example:

Input: 123123
Output: Mirror Number

Input: 123456
Output: Not a Mirror Number

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8. Coin Change Counter

Problem Statement:
Given an amount, determine the minimum number of notes required using:

₹500, ₹200, ₹100, ₹50, ₹20, ₹10

Example:

Input: 880

Output:
500 x 1

200 x 1
100 x 1
50 x 1
20 x 1
10 x 1

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9. Suspicious Transaction Detector

Problem Statement:
Input transaction amounts continuously.

Stop when -1 is entered.

Count:

- Transactions above ₹50,000
- Transactions below ₹1,000
- Total transaction amount

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10. Lift Movement Simulation

Problem Statement:
A lift starts at floor 0.

The user repeatedly enters destination floors.

Display:

- Floors travelled in each trip
- Total floors travelled
- Stop when user enters -1

Example:

Current Floor: 0
Enter Destination: 5

Travelled: 5 floors

Enter Destination: 2

Travelled: 3 floors

Total Travelled: 8 floors